Chapter 40 Using power handling

This chapter describes **Hunter 16's** sophisticated power monitoring, charging, conservation and control systems. These have no equivalent on a desktop computer.

40.1 The facilities

Hunter 16 has been designed for convenient and efficient operation, and its sophisticated power handling is a very important part of this.

Hunter 16's power-related facilities fall into the following main categories:

- Power-up
- Power-down
- Power monitoring and charging
- Saving power
- Power output

40.1.1 Power-up

Any of the following can cause **Hunter 16** to power-up:

- The user pressing a power-up key
- The RTC alarm going off
- Receiving RI (eg from an autoanswer modem)

Each cause can be enabled or disabled independently under program control. A program can also determine the reason for power-up, so it can respond appropriately. For example, a program can conserve power by turning off (see the next section) while waiting for a call or the correct time. (See section 40.1.4 for more information on saving power.)

40.1.1.1 Power-up state

When turned on, **Hunter 16** will normally resume execution from the state at which it was turned off. This is called 'resume mode', and helps make portable operation extremely convenient.

You can cancel resume mode by using INT 15H, function 64H, and **Hunter 16** will behave more like a desktop computer, rebooting whenever it is turned on. (Resume mode will also be cancelled if all battery power has been exhausted, including the backup battery.)

The backlight and contrast levels will normally power up set to values suitable for normal operating conditions, but you can set their power-up states by using INT 15H functions 74H and 75H, respectively.

If the user has removed the battery cap, **Hunter 16** will normally display the power options HOT key menu when next turned on, to allow the user to specify if the batteries have been changed. You can disable this facility by using INT 15H, function 7EH (this does not stop the user selecting the HOT key manually).

40.1.1.2 Power-up keys

The main power-up key is [PWR]. This key cannot be disabled or redefined; if there is any power in the batteries, it will always turn Hunter 16 on (although Hunter 16 may turn off again immediately, for example if the battery power is beneath the low power level).

The easiest way to check if [PWR] has caused the power-up is to enable the power key event (using INT 15H function 6EH) before power down, then check the event details on power up, using INT 15H function 6DH.

You can also define other power-up keys, by using INT 15H, function 71H. If you have done this, to determine if one of these is the reason for power up, you must enable the other possible power-up events (RI and alarm) before power down, then check if either is active after powerup; if not, then a power-up key must have been pressed.

40.1.1.3 RI

RI is an input on pin 22 of the 25-way D type (COM1), which will normally power up Hunter 16 if it goes high. You can disable power-up by using INT 14H function 26H. NOTE: If you are connecting to the RI input, do not leave it permanently high, as this will interfere with normal operation.

One way to check if RI has caused the power up is to enable the RI event (using INT 15H function 6EH) before power down, then check

the event details on power up, using INT 15H function 6DH. (Another way is to check the modem status register, using INT 14H function 3.)

40.1.1.4 RTC alarm

The RTC alarm is set by using INT 1AH, function 6. When the RTC time matches the RTC alarm time, it will turn Hunter 16 on. You can disable the alarm by using INT 1AH function 7.

The easiest way to check if the alarm has caused the power up is to enable the alarm event (using INT 15H function 6EH) before power down, then check the event details on power up, using INT 15H function 6DH. (Another, more laborious, way is to read the RTC time and alarm time, using INT 1AH functions 3 and 9, and compare them.)

40.1.2 Power-down

While turned off, **Hunter 16** will normally remember the current execution state so that it can restore it when turned on again. This is called 'resume mode'. Resume mode can be disabled by using INT 15H, function 64H, and will be cancelled if all battery power has been exhausted (ie including the backup battery).

Hunter 16 can power down when:

- The user presses the [PWR] key.
- The battery power falls below the low power threshold (in
- Advanced Power Mode only)
- The keyboard has not been touched for a period exceeding the
- idle timeout.
- The power-down function (INT 15H, function 68H) is used.
- The battery voltage falls below 4.5V.

With the exception of the last condition, all these causes can be enabled, modified, or disabled under program control. This allows programs to ensure that crucial tasks (such as transmitting or saving a file) are completed before power-down.

40.1.2.1 Power-down key

Pressing the [PWR] key will normally turn **Hunter 16** off, but this can be prevented by using INT 15H function 69H. (This function also prevents power down by the power-down function and idle timeout.)

40.1.2.2 Low power

In Advanced Power Mode (see section 40.1.3) **Hunter 16** will normally display low power warnings every 5 seconds and turn off automatically after 10 warnings, if the power level falls below the low power threshold (5%). This provides a large safety margin, by making sure that sufficient battery power remains to protect the memory contents (including the RAM disk and system configuration) for several weeks.

You can change the threshold and interval between warnings by using INT 15H function 62H, and prevent the automatic power down by using INT 15H function 7DH.

The battery level is monitored automatically during charging and discharging of NiCads and during discharging of Alkalines. You can also set the battery level by using INT 15H function 66H; this will normally be used to set the level when changing batteries. You can set the battery type and total capacity by using INT 15H functions 7FH and 80H.

40.1.2.3 Idle timeout

To conserve power, **Hunter 16** will normally turn off automatically if left idle (no keyboard input or communications activity) for 5 minutes. When the user is ready to continue, he or she can simply press [PWR] to carry on as if nothing had happened.

You can change the timeout period (or disable it entirely) by using INT 15H function 63H. You can prevent the power down by using INT 15H function 69H (which also prevents power down due to the power-down key or function).

40.1.2.4 Power-down function

The power down function (INT 15H function 68H) can be used to turn off Hunter 16. Use it, for example, to save power when a task is complete, or if Hunter 16 simply needs to wait for the RTC alarm or an incoming call to turn it on (see section 40.1.1).

This function can be disabled by INT 15H function 69H.

40.1.2.5 Controlling power- down causes

INT 15H function 69H will prevent power-down due to the idle timeout, power-down key or power-down function. Use it, for example, to make sure that the current task is completed before power-down.

Note that this function cannot prevent power-down due to low power (see section 40.1.2.2) or loss of power.

40.1.2.6 Power loss

If the main batteries are removed or fail (voltage falls below 4.5V), or the AC adaptor is turned off and batteries aren't fitted, **Hunter 16** will immediately turn off regardless of the current activities, and will not turn on again until main power is restored.

When main power is restored, you should find that the memory contents, RAM disk files, etc, are all intact and that any program which was running when power was lost will continue executing. This is because **Hunter 16** has a backup battery fitted internally, charged from the main power source, which is sufficient to maintain memory contents for at least 4 weeks.

If main power is lost for longer than 4 weeks, data corruption may occur.

40.1.3 Power monitoring and charging

Hunter 16 provides two distinct power monitoring modes: Standard

Power Mode and Advanced Power Mode. You can select the power mode by using INT 15H function 89H and users can change the mode by using the HOT key Power option.

To prevent users changing power modes or other power settings via the HOT key, use INT 15H function 7BH (to disable the HOT key) or 7CH (to disable the Power option). You can also replace the Power option routine with your own, by using INT 15H function 85H.

Charging is normally disabled outside the temperature range 5 to 40 degrees Centigrade but you can change the range by using INT 15H function 8BH.

For operating details, such as changing batteries and using the AC adaptor, see Part 2.

40.1.3.1 Standard Power Mode (SPM)

In Standard Power Mode, **Hunter 16** monitors the battery voltage to detect the low power condition. Charging is normally enabled at 70mA, but you can disable charging by using INT 15H function 8AH and enable fast (boost) charging at 200mA for an hour by using INT 15H function 8DH.

40.1.3.2 Advanced Power Mode (APM)

In Advanced Power Mode, **Hunter 16** monitors power consumption and recharging, so it can calculate the current charge level and help prevent problems with running out of power at a crucial time. This charge level can be read by users (by pressing the HOT key) and programs (using INT 15H function 61H), as a percentage of full charge. Charging is regulated automatically, starting at 200mA and then reducing to a trickle charge (about 10mA) when the batteries are fully charged.

Power level monitoring relies on measuring the current drain during use and the current supplied while charging NiCads. The battery level is thus calculated by subtracting from or adding to the current level.

For this to work, **Hunter 16** must know four other things:

- The initial charge level (typically 0% for new NiCads, 100% for new Alkaline batteries, but may be another figure if part-used batteries are fitted).
- The charging authorisation, which specifies whether charging can be selected, and for how long. (Attempting to charge Alkaline batteries is futile and hazardous.)
- The battery type, which specifies whether charging is allowed now.
- The battery capacity (NiCad capacity is much lower than Alkaline capacity).

This information can be provided by using the power option of HOT key, as described in Part 2. Hunter 16 will also prompt for it whenever you turn on after changing batteries. This is very convenient if the unit will only be used by responsible personnel, but is open to abuse by less careful users. For example, a user might be tempted to lie about the power level if they are getting low power warnings, risking total power loss.

If you cannot trust users with these facilities, you can disable them by using INT 15H functions 7CH and 7EH. You could then provide supervised, restricted access from within the application by using other INT 15H functions described below. Alternatively, you could re-enable the HOT key (using function 7CH) if the user types the correct password, to allow qualified personnel to recharge and swap batteries quickly and easily.

40.1.3.3 APM: Battery power level

To set the power level, use INT 15H function 66H. This should not be necessary when using NiCads, except when fitting a fresh set, but should be used whenever Alkaline batteries are changed, as **Hunter 16** has no way of knowing the new batteries' power level.

To read the power level, use INT 15H function 61H. This returns a value representing the percentage charge remaining. Remember that this indicates a different charge (and operating time), depending on the battery capacity.

40.1.3.4 APM: Charging authorisation

By default, in APM **Hunter 16** does not allow charging from the AC adaptor; it simply powers **Hunter 16** while connected. This is to avoid the risk of attempting to recharge Alkaline batteries.

Once NiCads have been fitted, use INT 15H function 65H to grant charging authorisation, then INT 15H function 7FH to enable charging.

You should typically only grant level 2, which allows charging until the user changes the batteries (or removes and replaces the battery cap), as this will stop users trying to charge Alkaline batteries. Charging must be carefully controlled; see section 40.2.3 for an example.

40.1.3.5 APM: Battery capacity

To change the battery capacity, use INT 15H function 80H. You should only need to do this if the user fits a different type of battery. The capacity should be specified by the manufacturer, but a typical NiCad has a capacity of 600-800mAh and a typical Alkaline battery has a capacity of about 1400mAh.

40.1.4 Saving power

To make the best use of its battery capacity, **Hunter 16** conserves power in a variety of ways, some obvious, some not so obvious. Most of these mechanisms are entirely automatic and have very little impact on users or programming, so they need not concern you. However, if you are writing programs with unusual requirements, you may need to know about the mechanisms described here.

The power-saving aspects fall into the following categories:

- Power modes
- Idle timeouts
- Switchable devices

40.1.4.1 Power modes

Hunter 16 can operate in a number of different modes, with different

power consumption and capabilities, as follows:

Mode	Description
Normal	Normal full speed operation at 8MHz
Slow1	Normal operation but at 4MHz
Slow2	Normal operation but at 2MHz
Halt	Execution stopped, waiting for RESET, NMI or other enable hardware interrupts
Stop	Execution stopped, waiting for RESET or NMI only

While turned on, **Hunter 16** is usually in NORMAL mode, but switches to STOP mode whenever it is waiting for keyboard input, to conserve power; when a key is pressed, it switches back to NORMAL mode.

As STOP mode does not respond to interrupts other than RESET and NMI, **Hunter 16** can miss serial data while waiting for keyboard input, if a communications package which relies on interrupts (such as Husky communications) is selected.

To reduce this risk but still save power, **Hunter 16** will use HALT mode instead of STOP mode, once an application has started communications (ie used INT 14H function 1 or 2).

If you think that there still might be problems, you can force **Hunter 16** to use HALT mode instead of STOP mode, by using INT 15H function 67H. This will make sure that no interrupts are missed, but uses a little more power.

Normally, if an application is using BIOS services **Hunter 16** will remain in NORMAL mode even if there is no user input. To allow use of low power modes even while certain services are being used, use INT 15H function 8CH.

You can also select the two 'slow' modes directly by using INT 15H function 60H, to conserve power at the expense of processing speed. For example, you might select the slowest mode while waiting for an Occurrence that cannot be used to turn **Hunter 16** on (as described in section 40.1.1).

40.1.4.2 Timeouts

Hunter 16 implements timeouts to conserve power when devices have been idle for a period. These are:

- The keyboard idle timeout. This turns **Hunter 16** off if the keyboard is not touched for the timeout period and communications is not active (buffers are empty). To change or disable this timeout, use INT 15H function 63H.
- The backlight idle timeout. This turns the backlight off if the keyboard is not touched for the timeout period. To change or disable the timeout, use INT 10H function 65H.

40.1.4.3 Switchable devices

There are two devices which consume significant power, that you can control from programs. You should keep their use to a minimum:

- The backlight. Use INT 10H function 64H to set the level, and function 65H to set a timeout.
- The RS232 interface. This turns on automatically when needed, but programs should turn it off when not needed, See section 38.2.4 for details.

40.1.5 Power output

Pin 10 of the 25-way D type provides a power output which may be used for operating low power devices. This can provide 5V +0.5/-1.0V at up to 100 milliamps, while **Hunter 16** is switched on. To control this, use INT 15H function 81H.

40.2 Using the facilities

40.2.1 Setting up

Hunter 16 allows you to use a range of power sources and provides a wide range of user and programming facilities related to their use. This section provides a few simple guidelines on setting up units to use either alkaline or NiCad batteries, which should provide a basis for your procedures. You will probably need to modify your procedures in the light of experience.

Switching units from one battery type to the other requires a little more care. This is described in Part 2, and should normally only be undertaken by reliable personnel, typically within your technical support department.

40.2.1.1 Using alkaline batteries

If the users are reliable, simply familiarise them with the instructions in Part 2, on changing batteries and using the HOT key and provide them with spare batteries. This approach is preferable since it involves you in the least extra work!

If the unit will be used by non-technical operators exclusively with alkaline batteries, set it up as follows:

- Use the HOT key power option (see Part 2) to:
    - Set the power remaining to 100% (if fresh batteries)
    - Set the low power warning onset to 5%
    - Set battery chargeable to No
    - Set battery capacity to 1400mAh (or as indicated by the
    - manufacturer)
    - Set charging authorisation to 1

    - Leave the HOT key.
- Create a simple utility which:
    - Inhibits the HOT key power option (INT 15H function 7CH)
    - Inhibits the battery change power option (INT 15H function 7EH)
- Run the utility on each unit.
- In the application, provide a battery change option (perhaps protected by a password), which resets the power level to 100% (INT 15H function 66H).
- Install the application and give out the units fitted with alkaline batteries (plus spares). Do not provide AC adaptors or NiCads.
- Tell users about the low power warnings and how to use the HOT key to check the power level, so they will know when to change the batteries. Tell them how to use the battery change option in the application.

If users do not need to be able to change batteries while away from home base, you need not provide the 'battery change' option or the spare batteries; instead, you could provide all battery changing facilities yourself.

40.2.1.2 Using NiCads

If the users are reliable, simply familiarise them with the instructions in Part 2, on recharging batteries and using the HOT key, and provide them with AC adaptors. This approach is preferable since it involves you in the least extra work!

If the units will be used by non-technical operators exclusively with NiCads, set each on up as follows:

- Fully charge the batteries and fit them to the unit.
- Use the HOT key power option (see Part 2) to:
    - Set the power remaining to 100%
    - Set the low power warning onset to 5%
    - Set charging authorisation to 2
    - Set battery chargeable to Yes

    - Set battery capacity to 800mAh (or as indicated by the manufacturer)
    - Leave the HOT key.
- Create a simple utility which:
    - Inhibits the HOT key power option (INT 15H function 7CH)
    - Inhibits the battery change power option (INT 15H function 7EH)
- Run the utility on each unit.
- Install the applications and give out the units fitted with NiCads and provide AC adaptors. Do not provide Alkaline batteries.
- Tell users about low power warnings and how to use the HOT key to check the power level, so they will know when to recharge. Make sure they know how to recharge. Tell users not to remove the battery cap (as this will disable charging).

If users do not need to be able to recharge batteries while away from home base, you need not provide the AC adaptor; instead, you could provide all battery charging facilities yourself. For example, you could arrange to recharge all units overnight, at the end of each day.

40.2.2 Using SLOW mode

The slow modes simply reduce the clock speed, saving power but making programs run slower! Their main use is to conserve power when execution speed is not important. For example, when waiting for something to process, or when performing tasks which require very little processing.

For example, suppose that **Hunter 16** is connected to a modem which does not provide RI and the application needs to store data received via the modem. You cannot turn **Hunter 16** off and wait for RI (as described in section 38.3.3), as the modem doesn't provide it.

For example, you could use the following code to minimise power consumption:

```
START:
select slow mode 2 (INT 15H, 60H)
```

```
LOOP1:
check modem status (INT 14H, 3)
if no DCD goto LOOP1

select normal mode (INT 15H, 60H)
receive and store file

goto START
```

If you can arrange for **Hunter 16** to turn on when it needs to do some Processing, you can conserve even more power by turning **Hunter 16** off! For example, if you simply need to respond at regular intervals, you can set the RTC alarm for the neXt time and that will power-up **Hunter 16**. If the event that requires processing can provide a suitable signal for connecting to the RI input, you can use that to power-up **Hunter 16** (see section 38.3.3).

40.2.3 Charging NiCad batteries

By default, users can authorise charging by using the power option of the HOT key, but you should disable this (INT 15H functions 7CH and 7EH) if you think there is any likelihood that they will fit Alkaline batteries and then attempt to charge them. REMEMBER, ATTEMPTING TO RECHARGE ALKALINE BATTERIES IS HAZARDOUS!

If you do disable the HOT key power option, you will need to provide charging facilities to someone. For example, you might perform all recharging yourself, in which case you could write a simple utility which grants temporary access to the HOT key, but not give copies to users.

Another method would be to build recharging facilities into each application or into a separate utility, but protect them by a password, which you only provide to responsible individuals. The following examples show how to implement either solution, together with a number of useful refinements.

40.2.3.1 Battery charging utility

The following code shows how you might implement a battery charging utility, which allows you to check the battery charge and time to recharge, or alter the battery power level.

The utility turns **Hunter 16** off to minimise the charging time and uses the RTC to turn it on again when charging should have finished. When **Hunter 16 is** turned on again (by the alarm or the user pressing [PWR]), the utility checks whether charging has finished, and if not, gives the user the option of carrying on charging, or returning to DOS.

Note that the utility only grants charging level 2, so that when **Hunter 16** turns on again, charging is disabled again. This means that the user must type the password again each time **Hunter 16** turns on, to continue charging.

```
;****************************
;* EXAMPLE: BATTERY UTILITY *
;****************************

START:
disable HOT key power option (INT 15M, 7CH)
disable battery change power option (INT 15H, 7EH)

CHARGE:

get charge level (INT 15H, 61H)
calculate operating time left (level * 14/100]
calculate recharge time [(100-level) * 12/100]
display charge level, operating time, recharge time

prompt "To recharge or alter level, type password"
get password
if password invalid, exit to DOS

prompt "Type new charge level or just ENTER"
if new value entered
     set power level (INT 15H, 66H)
     goto CHARGE                   ;SO CAN USE NEW POWER LEVEL


prompt "Are you sure NiCads are fitted? IF UNSURE, CHECK NOW
if reply is no, exit to DOS

tell user to connect adaptor
set alarm to [time now + recharge time)
remind user when will be fully charged
delay 30 seconds (for user to reed message)
```

```
CHARGING:
select charging level 2 (INT 15H, 65H);CHARGE UNTIL CHANGE
BATTERIES
enable charging (INT 15M, 7FH)     ;TO MINIMISE RECHARGE TIME
power down

;POWERED UP BY PWR KEY OR BY ALARM

display charge level
if level=100%
     prompt to disconnect adaptor
     delay 30 seconds (for user to read message)
     exit to DOS
else
     goto CHARGE
```

40.2.3.2 Battery charging within an application

The following code illustrates how you might implement control from within an application. This is quite similar to the charging utility, but as the user is using an application, the code gives the user the option to carry on using **Hunter 16** while it is recharging. This requires two different estimates of the recharge time, and different logic.

```
;*************************************************
;* EXAMPLE: BATTERY CHARGING IN AN APPLICATION *
;*************************************************

START:

disable HOT key power option (INT 15H, 7CH)
disable battery change power option (INT 15H, 7EH)

;MAIN PROCESSING LOOP


if user selects recharge option
     call CHARGE
get power level (INT 15H, 61H)

if power < LOW POWER
     call CHARGE    ;LOW POWER VALUE MUST BE MORE
                    ;THAN LOW POWER THRESHOLD
```

```
if ONCHARGE and power=100%
     call ENDCHARGE


;END OF MAIN PROCESSING LOOP



;*********************************************************
;* CHARGER ROUTINE CONTROL CHARGING FROM AN APPLICATION *
;*********************************************************

CHARGE:

get charge level (INT 15H, 61H)
calculate operating time left (level * 14/100]
calculate off-recharge time ((100-level) * 12/1001
estimate on-recharge time t(100-level) * 30/100)
display charge level, operating time, on-recharge time, off-
recharge time

prompt "To recharge, type password. To return, just type ENTER"
if password is invalid, return

prompt "Are you sure NiCads are fitted? IF UNSURE, CHECK NOW"
if reply is no, return

tell user to connect adaptor

prompt "Do you want to carry on using application while
charging?"
if reply is yes
     call ONCHARGE
else call OFFCHARGE           ;TO MAIN PROGRAM
return

;**********************************************************
;* OFFCHARGE ROUTINE - CHARGING WITH HUNTER 16 TURNED OFF *
;**********************************************************

OFFCHARGE:
set alarm to [time now + off- recharge time]
remind user when will be fully charged
delay 30 seconds (for user to read message)
authorise charging level 2    ;CHARGE UNTIL CHANGE BATTERIES
enable charging
power down                    ;TO MININISE RECHARGE TIME

; POWERED UP BY PWR KEY OR BY ALARM
get power level (INT 15H, 61H)
display charge level
```

```
if level=100%                  ;CHARGING OVER
     prompt to disconnect adaptor
     delay 30 seconds (for user to read message)
     return                    ;FINISHED CHARGING
else
     goto CHARGE               ;SO CAN CONTINUE CHARGING/QUIT


;*********************************************************
;* ONCHARGE ROUTINE - CHARGING WITH HUNTER 16 TURNED ON *
;*********************************************************

ONCHARGE:
remind user when will be fully charged
delay 30 seconds (for user to read message)
authorise charging level 2    ;CHARGE UNTIL CHANGE BATTERIES
set ONCHARGE flag to ON
return


;*************************************************************
;* ENDCHARGE ROUTINE - END CHARGING WITH HUNTER 16 TURNED ON *
;*************************************************************

ENDCHARGE:
tell user fully charged
tell user to disconnect charger
authorise charger level 1     ;CHARGING DISABLED
delay 30 seconds (for user to read message)
set ONCHARGE flag to OFF
return
```